

7. Sexual method of propagation includes.....

- a) Seed propagation
- b) Root propagation
- c) Rhizome propagation
- d) None of these

8.is responsible for ripening of fruits.

- a) IAA
- b) Ethylene
- c) 6-BA
- d) Absciscic acid

9. The temperature required for the cultivation of Cinchona is..... °F.

- a) 45 - 60
- b) 60 - 75
- c) 75 - 90
- d) 90 - 105

10. Cultivation ensures.....

- a) Irregular supply of drug
- b) Quality of drugs
- c) Quality and purity of drug
- d) All of above

11. Water insoluble portion of gum tragacanth is known as.....

- a) Tragacanthin
- b) Bassorin
- c) Inotragacanthin
- d) Guranin

12. Gum acacia is obtained from which plant?

- a) *Acacia arabica*
- b) *Acacia urens*
- c) *Acacia ribosa*
- d) *Acacia amansii*

13. is used as carbon source in Plant tissue culture medium.

- a) Sodium citrate
- b) Agar
- c) Disodium EDTA
- d) Sucrose

14. Alkaloids gives coloured precipitate along with Mayer's reagent.

- a) Reddish brown
- b) Reddish orange
- c) Yellow
- d) Cream

15. Molecular formula for isoprene subunit is.....

- a) C_4H_8
- b) C_5H_8
- c) C_6H_8
- d) C_7H_8

16. As per the Unani system of medicine **Al-Nar** represents the.....

- a) Water
- b) Fire
- c) Air
- d) Earth

17.is the optimum pH range of culture medium for growth of plant tissue.

- a) 4.5 to 5.0
- b) 5.0 to 5.5
- c) 5.5. to 6.0
- d) 6.0 to 6.5

18. Borntrager's test is positive only in case of type of anthraquinone glycosides.

- a) -C-type
- b) -O-type
- c) -S-type
- d) -N-type

19. Concept of **Triguna** is described in which system of medicine?

- a) Ayurveda
- b) Unani
- c) Siddha
- d) Chinese

20. Sting of bees and hornets are example of _____ class allergen.

- a) Ingestant
- b) Injectant
- c) Infectant
- d) Inhalant

Q.II. Answer any TWO of the following.

[10x2=20]

1) Differentiate between primary metabolites and secondary metabolites. What are alkaloids? Explain chemical classification of alkaloids with suitable examples and Write various preliminary phytochemical tests for detection of alkaloids.

2) Define plant tissue culture. Write detail note on composition of culture media. Enlist and explain various types of cultures. Write note on applications of plant tissue culture in Pharmacognosy.

3) What is the need of classification of crude drugs? Illustrate Alphabetical, Morphological, Chemical and Pharmacological classification of crude drugs.

Q.III. Answer any SEVEN of the following.

[05x07=35]

1) Describe in detail various sources of crude drugs and differentiate in between organized and unorganized crude drugs.

- 2) What do you mean by plant growth regulators? Explain various plant growth regulators along with suitable examples.
- 3) Discuss in detail factors affecting cultivation of medicinal and aromatic plants.
- 4) Illustrate theory and basic concepts, diagnosis and treatment aspects of Ayurvedic system of Medicine. Add note on role of pharmacognosy in alternative system of medicine.
- 5) Classify resins and tannins with suitable examples and give chemical tests for detection of glycosides.
- 6) What do you mean by edible vaccines? Enlist differences between conventional vaccines and edible vaccines? Add note on methods of transformation and mode of action of edible vaccines.
- 7) Describe in detail microscopic methods of evaluation of crude drugs.
- 8) Give biological source, method of preparation, chemical constituents, chemical tests and uses for following drugs
 - a) Agar
 - b) Acacia
- 9) Give the broad classification of marine drugs and explain each class in detail with suitable example.

●●●Paper End●●●